

Climate, Migration, and Regional Development

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Index

1. Introduction
2. Data
3. Identification
4. Results
5. Further Work

Index

1. Introduction
2. Data
3. Identification
4. Results
5. Further Work

Being born is the single most important lottery you play

- Location at birth is largely a roll of the dice
- Rolling bad on climate yields a lifelong disamenity
- Improving local climate is extremely costly
- Migration is thus the main escape mechanism

Key Question

Do climate disamenities make people respond more strongly to reductions in migration costs?

And if so, **does this drive regional economic development?**

Contribution

Climate migration events and their consequences have been studied far and wide

- Disasters may lead to permanent migration (Belasen & Polacheck 2013) or to temporary flows (Stanley 2010)
- Plain bad climate may also drive moves (Saldana 2009; Halliday 2006)
- Model implied welfare measures point to reasonable-to-large gains (Desmet & Rossi-Hansberg 2020; Morten & Oliveira 2025; Kennan & Walker 2008)
- As do measurements of individual level income gains (Bryan & Mobarak 2014; Hornbeck 2002)

This project: Are migrations responses to costs also amplified? And, can we get a more direct aggregate welfare measure?

Laterally, but important to me: better understanding regional development stories in Brazil

Findings

The main preliminary results are that:

- **Climate:** No robust effect on migration responses
- **Welfare:** Migration leads to positive, sizable GDP per capita gains
 - I find a 2.3% increase in GDP per capita attributable to migration
 - The closest analogue, Morten-Oliveira, find a 0.6% total welfare effect
- Both results seem robust

Caveat: Data limits the amount of controls available for the welfare result

Index

1. Introduction
2. Data
3. Identification
4. Results
5. Further Work

Data at a Glance

- **Migration:** 1980–2010 (mesorregions); 1940–2010 (state), decennially from censuses, via Morten Oliveira (2025)
- **Climate conditions:** 1960s onward, daily from BR-DWGD
- **Transport (travel time):** 1940–2010, decennially from Morten-Oliveira (2025)
- **GDP & Population:** 1940–2010, decennially (mostly) from IPEA
 - GDP per capita: a model agnostic welfare measure

Constraints: State-level data has few regions; mesorregion has few years. Climate data is unavailable for pre Brasília periods

GDP and Population Summary Data Summary

Table: Summary Statistics: GDP and Population Growth

Variable	N	Mean	σ	Min	P25	Median	P75	Max
GDP per capita growth	2,857	0.03	0.06	-0.42	-0.01	0.03	0.06	0.74
Population growth	2,859	0.02	0.02	-0.13	0.01	0.01	0.02	0.37
GDP growth	2,857	0.05	0.07	-0.39	0.01	0.05	0.09	0.75

Notes: Sample covers 1920–2010 for 137 mesoregions (fewer regions available in early decades). All variables are annualized log changes.

Migration and Travel Time Data Summary

Table: Summary Statistics: Migration Flows and Travel Time

Variable	N	Mean	σ	Min	P25	Median	P75	Max
Migration flows	74,528	459	2,304	0	0	24	128	89,896
Log travel time absent roads	74,528	4.85	0.74	1.27	4.44	5.03	5.41	6.00
Log travel time on roads	74,528	3.24	0.88	1.03	2.69	3.24	3.71	5.48

Notes: Sample covers 1980–2010, decennial observations for origin–destination mesoregion pairs. Migration flows are in levels; travel times were obtained via a fast-marching algorithm (Morten & Oliveira, 2025).

Climate Data Summary

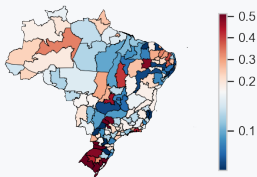
Table: Summary Statistics: Climate Exposure Indexes

Variable	N	Mean	σ	Min	P25	Median	P75	Max
Composite exposure	2,466	0.23	0.49	0.00	0.04	0.12	0.21	6.77
Drought anomaly	2,466	0.00	1.00	-0.79	-0.55	-0.42	0.17	7.77
Drought absolute	2,466	0.00	1.00	-0.55	-0.47	-0.44	-0.40	5.01
Heat exposure	2,466	0.00	1.00	-0.54	-0.36	-0.32	-0.22	9.89
Flood exposure	2,466	0.00	1.00	-0.61	-0.49	-0.40	-0.29	8.24

Notes: Original data are daily observations from 1961–2024 at $0.1^\circ \times 0.1^\circ$ resolution, including precipitation, temperature (min/max), relative humidity, and evapotranspiration. Variables shown are decennial averages of standardized exposure indexes aggregated to the regional level.

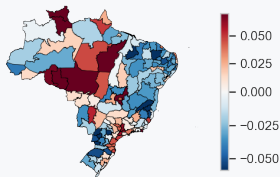
Spatial Patterns

Mean Extreme Climate Exposure
1970-2010



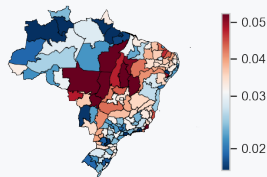
(a) Climate exposure

Net Migrant Receiving Rate
1980-2010, share of population



(b) Migration

Annualized GDP per capita Growth Rate
1970-2010



(c) GDP per capita growth

Index

1. Introduction
2. Data
3. Identification
4. Results
5. Further Work

Central Identification Idea

I leverage the Morten-Oliveira instrument to obtain an exogenous shock to travel times

Climate:

- Use transport shocks to identify migration responses
- Caveat: FEs capture region specific effects
 - Test: do people in high-exposure regions respond more strongly to the opportunity to leave?

Welfare:

- Recover migration induced by transport shocks
- This should be exogenous to regional welfare outcomes
 - Test: does net migration drive local welfare?

The welfare channel

GDP per capita is a good measure of welfare, but is codetermined with migration flows
⇒ the Brasília instrument allows for exogenous variation of migration!

Key issue: controls

- Exogeneity does not mean absence of omitted variables
- Road construction itself could drive GDP per capita outcomes
- I implement a gravity theory inspired market access control

Index

1. Introduction
2. Data
3. Identification
- 4. Results**
5. Further Work

Replication of the Morten-Oliveira Instrument

Table: Effect of Brasília-Induced Travel Time Reduction on Travel Time on Roads

	<i>Dependent variable: log(fm_road)</i>
Brasília Induced TT Reduction $\times$ 1940	-0.023*** (0.005)
Brasília Induced TT Reduction $\times$ 1960	-0.210*** (0.005)
Brasília Induced TT Reduction $\times$ 1970	-0.378*** (0.005)
Brasília Induced TT Reduction $\times$ 1980	-0.575*** (0.005)
Brasília Induced TT Reduction $\times$ 1990	-0.544*** (0.005)
Brasília Induced TT Reduction $\times$ 2000	-0.545*** (0.005)
Brasília Induced TT Reduction $\times$ 2010	-0.540*** (0.005)
Observations	149,056

Climate interaction draws a null

Table: Climate Exposure and the Effect of Instrumented Travel Time on Migration

	<i>Dependent variable: Log migration flow</i>		
	Baseline	Climate Exposure	Exposure Dummy
IV time travel	-1.432*** (0.044)	-1.420*** (0.046)	-1.432*** (0.044)
Climate exposure $\times$ IV tt		-0.064 (0.050)	
Exposure Dummy $\times$ IV tt			-0.010 (0.097)
Observations	43,640	43,640	43,640

Notes: Observations are origin–destination–year. Origin $\times$ year and destination $\times$ year fixed effects are included. Standard errors are clustered at the origin–destination level.

Interpretation: Climate may matter in levels, not in responses

Migration drives GDP and GDP per capita

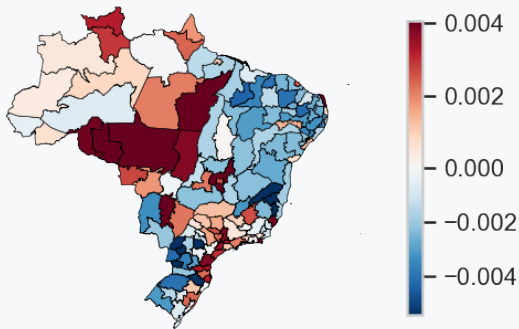
Table: Migration-Induced Growth Effects

	GDP per capita growth		GDP growth	
	RE	FE	RE	FE
Brasília induced net migrant receival rate	0.046*** (0.011)	0.081*** (0.018)	0.203*** (0.014)	0.224*** (0.023)
Market access gained	0.000 (0.000)		0.000 (0.000)	
Observations	547	547	547	547
Log-likelihood	1271.970	—	1196.057	—

Notes: Dependent variables are annual log growth in GDP per capita and GDP, respectively. Intercepts and year coefficients are omitted. Random-effects (RE) specifications include year fixed effects and random effects for region and region-specific time trends. Fixed-effects (FE) specifications include region fixed effects, year fixed effects, and region-specific time trends. Standard errors in parentheses.

The map tells a pretty clear story

GDP per Capita Growth due Migration
1970-2010



Agricultural frontier and urbanized regions are winners, the Northeast is the big loser. What could be driving this?

- Selection in migration?
- Agglomeration effects? Which ones?
- Non-linear (S-curve) regional dynamics?

In aggregate, internal migration increased GDP per capita by 2.1% from 1970 to 2010

Index

1. Introduction
2. Data
3. Identification
4. Results
5. Further Work

Robustness Checks

I'm satisfied with the climate null result. A better defense of the GDP/Welfare result requires:

- Replication entire pipeline at state level
- Better controls for omitted variables

The current setup allows for exogenous migration, but roads impact GDP in many other ways

Interpretation

Provided the GDP result is convincing, there are many ways to explore what drives it

- Selection: census microdata comparison of types of migrants
- Agglomeration: move away from the buzzword. Why do cities feature agglomeration and other places do not?
- Human capital accumulation? Better matching?
- What about the agricultural frontier? Resource abundance? A model could be useful here.

Figuring this out could go a long way towards building a better understanding of Brazilian mid-to-late 20th century development!